



Newly developed QCM-DNA biosensor for SNP detection in small DNA fragments: A wine authenticity case study

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ABSTRACT

We propose a QCM-DNA biosensor for single nucleotide polymorphism (SNP) detection in samples of differing complexity. An optimized protocol is presented, focusing on parameters affecting probe immobilization and hybridization efficiency. Our results led to the implementation of thiolated probe reduction with TCEP, followed by immobilization in PBS buffer containing $MgCl_2$. The biosensor exhibited an enhanced specificity at 37 °C, achieving detection of single mismatches using synthetic targets. Using real samples, we applied the biosensor in a wine authenticity assessment context. The addition of dithiothreitol improved stability and reproducibility when testing wine DNA samples. The QCM-DNA biosensor was able to specifically detect complementary DNA in leaf and wine DNA samples, distinguishing between samples with two heterozygous mismatches. The biosensor solely depends on DNA extraction, basic instrumentation, and reagents, without requiring PCR or signal amplification strategies. Our findings show the biosensor potential for applications in wine authenticity assessment and other fields requiring complex analysis in undemanding settings.

1. Introduction

In various fields of application, the development of new analytical instruments has been directed at overcoming the existing limitations and/or complementing the standard methodologies for sample analysis. This has led to the emergence of biosensors as portable, low-cost, and simple devices, that are able to produce a rapid response for a more practical analysis performed inside and outside laboratory settings (Lino et al., 2022; Tetyana, Morgan Shumbula, & Njengele-Tetyana, 2021).

Biosensors are devices that generate a measurable signal upon biorecognition of a target analyte, i.e. a molecule or substance of interest for detection (Turner, 2013). The two main elements in a biosensor are the bioreceptor, which is a biological element that can recognize and capture the analyte, and the transducer, that is responsible for converting the biorecognition event into a measurable signal, usually proportional to the detected analyte-bioreceptor interactions. Besides these elements, there are other electronic components that constitute a biosensor, whose role is to amplify and convert the transducing signal into digital form. A display unit is then responsible for generating a readable output for the

user in a suitable form (Naresh & Lee, 2021).

There are currently a wide variety of biosensors, classified based on the components used in their configuration. According to the main transducing principles applied, these can be classified as electrochemical, optical, and gravimetric (Yang, Ma, Stanciu, & Wu, 2020). There are advantages and limitations related to every type of biosensor and the choice of configuration will always depend on the intended use for the device (Vigneshvar, Sudhakumari, Senthilkumaran, & Prakash, 2016).

Among the various types of biosensors, gravimetric biosensors based on quartz crystal microbalances (QCM) are particularly interesting due to their increased sensitivity to mass changes in the nanoscale range. These devices are well-established and have demonstrated to be sensitive in detecting different analytes in different media (Alanazi, Almutairi, & Alodhayb, 2023; Lino et al., 2023). The physical structure of a QCM consists of a piezoelectric quartz crystal interposed between two metal electrodes, usually composed of gold. When the electrodes are subjected to an alternating electric field with a frequency close to the resonant frequency of the crystal, it vibrates vigorously and in a stable

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manner (Na Songkhla & Nakamoto, 2021). Upon mass deposition at the QCM surface, its resonance frequency decreases. Taking advantage of this mass loading effect on the registered frequency, a broad range of application was opened for QCM biosensors, based on the detection of biorecognition events at the QCM surface. These can be used for real-time and label-free detection of various biological and non-biological analytes, and are characterized by operation simplicity, sensitivity, affordable cost, and easy data analysis (Jandas, Prabakaran, Jingting, & Derry Holaday, 2021).

There is a wide variety of bioreceptors that can be employed in QCM biosensors. Among them, DNA aptamers have gathered a lot of attention, due to their easy and low-cost synthesis, enhanced biocompatibility, high specificity, higher thermal stability, and long shelf life (Yoo, Jo, & Oh, 2020). QCM-DNA biosensors have been developed for different applications in many areas, as in the clinical field (Lino et al., 2023), food safety analysis (Wachiralurpan, Chansiri, & Lieberzeit, 2020) and environmental testing (Kumar, Singh, & Verma, 2023). Target-specific aptamers can be synthesized and modified with functional groups, e.g. thiol, that allow their immobilization in the gold electrode present at the QCM surface, so it can be used to detect DNA present in a sample through complementary sequence hybridization (Hosseinzadeh & Mazloum-Ardakani, 2020).

Food authenticity assessment represents an application area where the development of DNA-based biosensors is less advanced, although they have vast potential as a fast, simple, and efficient analytical tool to identify the species, cultivar, or varietal origin of food products. Compared to other analytes that can be detected by biosensors, DNA is more stable and can withstand the harsh conditions of food processing. It is also abundant and can be easily extracted from most samples. These properties make DNA an ideal target for food authenticity testing and a suitable analyte for QCM-DNA based detection (Barrias, Ibáñez, Fernandes, & Martins-Lopes, 2024). These devices would be very valuable for the authenticity testing of numerous food products. One particularly interesting application for DNA biosensors is the varietal identification of wine, a product that belongs to a billion-euro sector and is a vulnerable target for fraudulent practices (Pereira et al., 2018). Although there are currently different molecular lab-based methodologies that can be applied for grapevine varietal identification (Barrias et al., 2023; Barrias, Ibáñez, & Martins-Lopes, 2024; Boccacci, Chitarra, Schneider, Rolle, & Gambino, 2020; Pereira et al., 2017), the establishment of biosensors can represent a more practical alternative tool to ensure authenticity throughout the entire wine chain. The aim of this work was to establish a simple functioning protocol in a QCM-DNA based biosensor for specific DNA detection in undemanding settings. To test its performance in real sample analysis, the biosensor was used to distinguish between two grapevine varieties at mismatch level, within the context of wine authenticity assessment.

2. Materials and methods

2.1. Probe and target design

All oligonucleotides used in this work were composed of 35 bp and were designed based on three single nucleotide polymorphisms (SNPs) identified in the *F3H* gene of *Vitis vinifera* L. (Gomes et al., 2018). The probe presented the following sequence: 5' – thiol – GCGAAAGCT-GAAGCTAATCTTTCTTTGTCTTTG –3'. Two non-complementary targets were designed differing in the number of mismatches present in their sequences (in bold and underlined), in relation to the probe: NC Target 3 MM (three mismatches), sequence 5' – CAAATACAAA-GAAAAGATTAGCTTCAGCCTA**ACGC** – 3' and NC Target 1 MM (one mismatch), sequence 5' –CAAAGACAAAGAAA**CA**TTAGCTTCAGCC TTTTCG – 3'. A complementary target (C Target) was designed to be fully complementary to the probe, with the sequence 5' – CAA AGA CAA AGA AAA GAT TAG CTT CAG CCT TTC GC – 3'. The oligonucleotides were acquired from Frilabo (www.frilabo.pt), were eluted in ultrapure

water to a stock solution of 100 μ M and further used at a concentration of 0.50 μ M in biosensor experiments.

2.2. Apparatus and electrode preparation

The biosensing platform included an OpenQCM Q⁻¹ device and AT-cut gold-coated quartz crystals suitable for detection in liquid media, all purchased from Novaetech SRL (Pompeii, Italy). The crystals had 14 mm of blank diameter, a gold electrode on the sensing surface with a diameter of 11.5 mm and a fundamental resonance frequency of 10 MHz. The device included a module with a 200 μ L capacity open chamber, allowing to directly pipette each sample onto the gold electrode at the surface of the crystal. Resonance frequency (Hz) was monitored in real-time using the Open QCM Q⁻¹ software. Before assembly, Au-coated quartz crystals were sonicated for 5 min in ethanol absolute $\geq 99.8\%$ (VWR Chemicals) and maintained in an incubator with controlled temperature set at 50 °C until completely dry.

2.3. DNA samples

Young leaves and monovarietal experimental wine samples from grapevine varieties Touriga Franca and Rufete were collected. Grapevine leaf samples were obtained from the certified vineyards Sogrape Vinhos S.A. and Real Companhia Velha, and immediately frozen in liquid nitrogen. Monovarietal wine samples were produced at the National Institute for Agricultural and Veterinary Research (INIAV), as described in Pereira et al. (2017) and stored at –20 °C. DNA was extracted from frozen leaf samples using the described CTAB method (Doyle & Doyle, 1987). DNA extraction from wine samples were performed as described by Pereira, Guedes-Pinto, and Martins-Lopes (2011). The DNA samples were eluted in 80 μ L of 0.1x Tris-EDTA (TE) buffer and maintained at –20 °C until use. Sample purity and quantity was determined on the Nanodrop™ 1000 Spectrophotometer. Before biosensor analysis, the concentration of all samples was adjusted to 0.50 μ M in 1x phosphate buffered saline (PBS) buffer.

2.4. Experimental design

After the individual crystals were installed in the QCM device, a calibration step was performed in air to determine the baseline resonance frequency for each crystal. After calibration, the measurement of the resonant frequency was started. First, the surface of the gold electrode was sequentially washed with distilled-demineralized water and 1x PBS (both autoclaved) by performing three pipetting steps with 200 μ L of each solution. After washed, 200 μ L of 1x PBS were introduced in the chamber, maintained until the temperature stabilised and was used to establish the baseline frequency for each experiment set. Four main steps were performed in each experiment: electrode functionalization, negative sample testing, positive sample testing and probe regeneration, with washing steps performed in between. This was done by performing three pipetting steps using only 1x PBS, after which 200 μ L of the same buffer were left on the chamber while frequency data was recorded for 15 min.

2.5. Electrode functionalization

Probe immobilization in gold electrodes is achieved through the interaction between gold and the thiol group at one end of the probe. The functionalization steps consisted in incubating 200 μ L of probe solution in the electrode surface for 30 min. Different 0.50 μ M probe solutions were prepared according to the protocol optimization experiment undertaken. For the immobilization of probe not subjected to any pre-treatment step, 1.5 μ L of probe stock solution was used directly in 1x PBS buffer. To obtain TCEP-Probe for immobilization, a 1:1 v/v dilution of probe stock solution with 20 mM tris(2-carboxyethyl) phosphine (TCEP) solution was performed, for a final concentration of

50 μM probe and 10 mM TCEP. The reaction was incubated at room temperature for 1 h. Then, 3 μL of TCEP-Probe were added in different immobilization buffers (IB): PBS (1x PBS); IB NaCl (1x PBS + 10 mM NaCl); IB MgCl_2 (1x PBS + 1 mM MgCl_2).

2.6. Hybridization and probe regeneration

The hybridization between complementary single DNA strands is at the basis of this biosensor detection principle. NC Targets, C Targets and DNA samples were used in the sample testing steps according to each experiment undertaken. To achieve hybridization, each sample solution was maintained in the biosensor surface for 15 min. To test the biosensor specificity for each assay, a NC Target or non-complementary DNA sample was first pipetted in the biosensor, before the C Target/complementary DNA sample. In every experiment, the hybridization buffer for all targets and DNA samples consisted in 1x PBS. Immediately before being pipetted into the chamber, the sample solutions containing the extracted DNA samples were denatured at 95 $^{\circ}\text{C}$ for 5 min, to guarantee the presence of single-stranded DNA, available to hybridize with the immobilized probe.

Following the C Target/complementary DNA sample hybridization step, the probe was regenerated by washing the assembled electrode with denaturation buffer consisting of 0.5M NaOH + 3M NaCl, to guarantee the availability of the probe. In each experiment, the hybridization/probe regeneration steps were performed in triplicate. The frequency recorded by the QCM device varied according to the mass deposited on the electrode surface. Fig. 1 schematizes the overall experimental design and the expected frequency variation trend.

2.7. Data analysis

Resonance frequency data was recorded during all experiments. The mass deposition resultant from probe immobilization and hybridization on the electrode surface was determined by calculating the frequency shift (Δf) for each step. For this, the first 300 data points acquired upon signal stabilization in each 1x PBS washing step were used and a mean value was determined. The frequency shift caused by probe

immobilization was calculated by subtracting the mean value of the baseline washing step (f_0) to the mean value of the 1x PBS washing step post-probe (f_1). The frequency shifts resultant from the hybridization steps were obtained by subtracting the 1x PBS washing step post-probe (f_1) to the 1x PBS post-hybridization washing step (f_{h1-2}). This was performed for the three replicates in each experiment and mean $\pm$ standard deviation (SD) was calculated. All statistical analysis was done through a T-student or one-way ANOVA test using the Statview Version 5.0.1 software to test for significance ($p < 0.05$).

3. Results and discussion

3.1. Effect of TCEP treatment on probe immobilization

Probe immobilization is a key step to develop a robust DNA-based biosensor. The probe acts as a bioreceptor, responsible for capturing the targeted analyte through specific hybridization. When probe immobilization on the transducer surface is not adequate, the detection most likely will not occur and the biosensor will not produce reliable results (Barrias, Ibáñez, Fernandes, et al., 2024). Immobilization of thiolated probes onto gold surfaces is at the basis of many biosensing platforms, most of which include a probe pre-treatment step with a reducing agent, namely TCEP (Hwu et al., 2020; Qin, Wang, Tang, Tian, & Wu, 2024; Xu et al., 2021).

The first experiments performed aimed to compare the effect of probe TCEP pre-treatment in the immobilization step and further hybridization efficiency of a QCM-DNA based biosensor (Fig. 2). The patterned grey bars represent the frequency shifts Δf (Hz) obtained throughout the experiment where the thiolated probe was immobilized without any pretreatment. Blue solid bars represent the Δf obtained throughout the experiment where the thiolated probe was pretreated with TCEP before immobilization.

The Δf resultant from probe immobilization was -17 ± 7 Hz, while the Δf resultant from TCEP-Probe immobilization was -41 ± 8 Hz (Fig. 2A). The higher Δf achieved for the pre-treated probe shows that a higher quantity of probe was immobilized on the electrode surface. This is caused by the TCEP induced reduction of the disulfide present at the

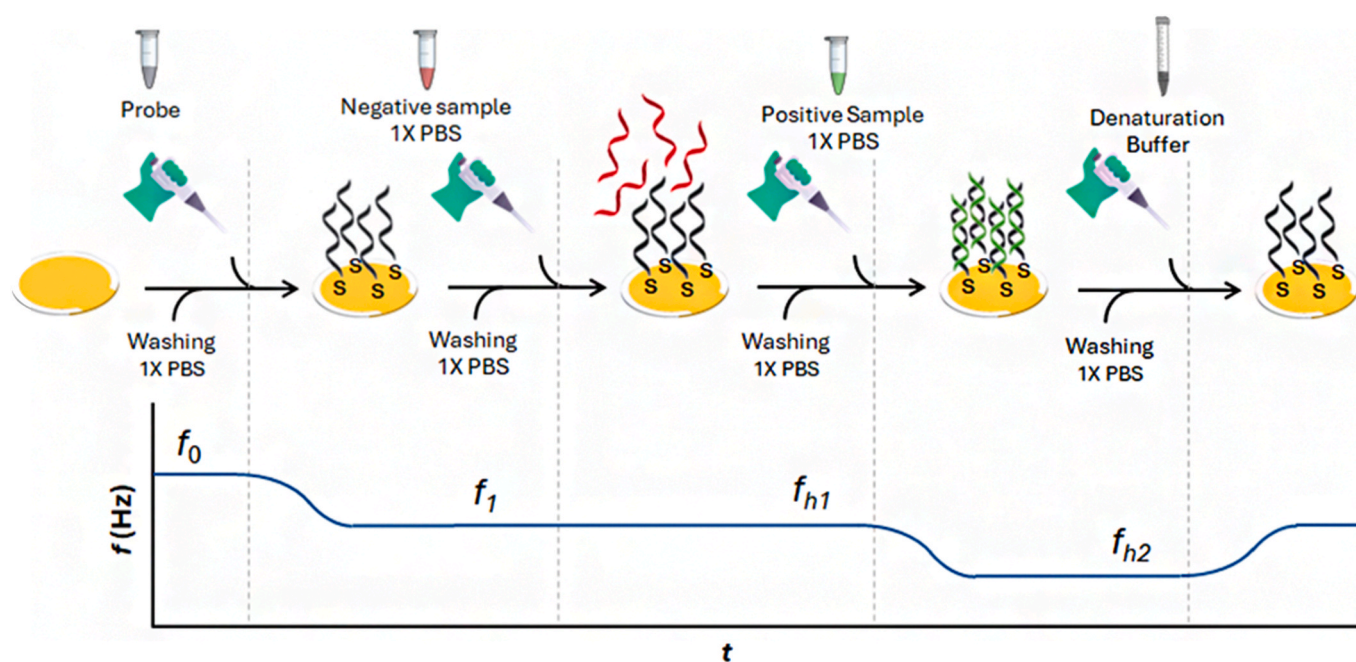


Fig. 1. Scheme of the overall experimental design and expected frequency [f (Hz)] variation trend through time (t). With several 1x PBS washings in between steps, each electrode was functionalized with probe, followed by sequentially testing negative and positive targets/samples and regenerating the probe after each hybridization. During the experiments, the frequency variation in relation to the deposited mass was recorded for further data analysis.

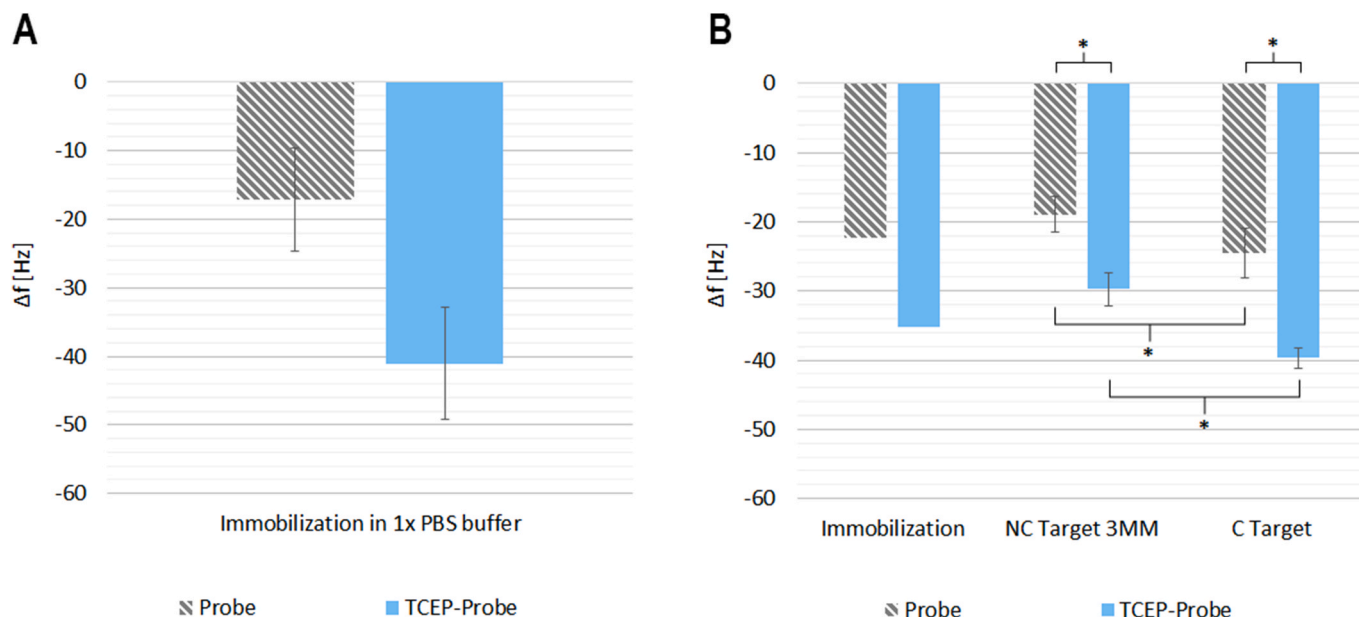


Fig. 2. Frequency shifts Δf (Hz) obtained in the experiments conducted to compare the effect of probe pretreatment in (A) immobilization and (B) hybridization efficiency. Patterned grey bars represent the Δf obtained throughout the experiment where the thiolated probe was immobilized without any pretreatment. Blue solid bars represent the Δf obtained throughout the experiment where the thiolated probe was pretreated with TCEP before immobilization. NC Target 3 MM was used as negative control whose sequence differed in three mismatches in relation to the positive control sequence (C Target). All hybridization steps were performed in 1x PBS. The error bars show the standard deviation considering three experiment replicates. *Statistically significant signal for a p-value <0.05.

probe end, into a thiol group that became available to form a covalent bond to the electrode gold surface, as described by [Javadi-Zarnaghi and Dastjerdi-Khorzoghi \(2019\)](#).

To test the effect of both probe immobilization strategies in the biosensor hybridization efficiency, two experiments were conducted in triplicate, adding two steps of hybridization in 1x PBS ([Fig. 2B](#)). First, a negative control (NC Target 3 MM) was tested. Considering the 35 bp sequence targeted by the designed probe, this target has three mismatches in its sequence and in ideal conditions it would not be able to hybridize with the probe, nor cause any significant Δf , and should be removed by the subsequent washing step performed after incubating for 15 min. Then, a second target was added (positive target – C Target) that is fully complementary to the probe. Hybridization should occur between C Target and the probe, and a Δf should be expected, approximate to that obtained for probe immobilization.

The results from the experiment performed after probe immobilization without TCEP pretreatment are represented by patterned grey bars in [Fig. 2B](#). Probe immobilization resulted on a Δf value of -20 Hz. For the hybridization steps, a Δf of -19 ± 3 Hz was observed for NC Target 3 MM, and -25 ± 4 Hz for C Target. In the same figure, solid blue bars show the results from the experiment performed after TCEP-Probe immobilization, that resulted on a Δf value of -35 Hz, followed by a Δf of -30 ± 2 Hz for NC Target 3 MM, and -40 ± 1 Hz for C Target.

In both experiments, a statistically significant ($p < 0.05$) Δf was observed for C Target in relation to NC Target 3 MM, and these shifts were similar to the shift obtained for probe immobilization, proving the high hybridization efficiency using both probes. The results show that although hybridization was observed between the NC Target 3 MM and the probe, indicating a lack of specificity, it did not saturate the probe immobilized on the electrode surface and it was partially still available for C Target detection. At this stage, only synthetic DNA sequences were being introduced in the biosensor, so the frequency shifts observed should not be attributed to deposition of contaminants or other non-specific molecules on the electrode surface. The weak bonds that could hypothetically form between non-hybridized target sequences lying horizontally on the surface of the electrode, would be disrupted during the post-hybridization washing step, so these are also not

responsible for the observed higher Δf .

In the TCEP-Probe experiment, higher Δf values ($p < 0.05$) were obtained for NC Target 3 MM and C Target in comparison to those obtained in the untreated probe experiment, suggesting an improved biosensor performance where a TCEP pre-treated thiolated probe was immobilized. TCEP reduction of the thiolated probe was implemented on the biosensor detection protocol and TCEP-probe was used on subsequent experiments.

3.2. Buffer effect on probe immobilization

Herne and Tarlov were pioneers in characterizing the immobilization of DNA probes onto gold surfaces ([Herne & Tarlov, 1997](#)). Among other meaningful contributions to gold surface functionalization, their experiments demonstrated that the optimization of the molecular density of DNA probes immobilized at gold surfaces greatly depends on the ionic strength of the immobilization buffer used. The ionic strength is determined by the concentration of ions present in a solution, i.e. Na^+ , Cl^- , K^+ , or Mg^{2+} . The presence of salts at higher concentrations increases the ionic strength of the buffer and enhances DNA probe adsorption on gold surfaces. High strength buffers such as high-salt PBS supplemented with NaCl or MgCl_2 , help to overcome the electrostatic repulsion forces between neighbouring probes (composed by negatively charged phosphate groups), and between the probes and the negative charges that are typically found on gold surfaces ([Liu, Canoura, Alkhamis, & Xiao, 2021](#)). On the contrary, low ionic strength immobilization buffers (e.g. TE) may lead to low probe densities upon immobilization on gold, and the occurrence of horizontal probe adsorption at the electrode surface, blocking it from proper probe immobilization and resulting in false signal readings ([Batistuti, Bueno, & Mulato, 2020](#)). Other authors have also reported that the cations present in the immobilization buffer can also affect probe binding rate to gold surfaces. [Koo, Sina, Carrascosa, Shiddiky, and Trau \(2015\)](#) indicate that a change from Na^+ to Mg^{2+} in a buffer, composed also by Cl^- , can induce a faster DNA probe adsorption on gold nanoparticles.

The following experiments in our study aimed to compare the effect of three different immobilization buffers in the overall biosensor

performance: immobilization buffer 1x PBS (IB PBS), immobilization buffer 1x PBS with 10 mM NaCl (IB NaCl) and immobilization buffer 1x PBS with 1 mM MgCl₂ (IB MgCl₂). Fig. 3 shows the results from the immobilization experiments using the three buffers, regarding three immobilizations on individual electrodes.

A Δf of -41 ± 8 Hz was observed for TCEP-Probe immobilization in IB PBS (solid green bar). For the immobilization in IB NaCl, a Δf value of -63 ± 18 Hz was observed (patterned grey bar), and Δf value of -47 ± 10 Hz resulted from the TCEP-Probe immobilization in IB MgCl₂.

Higher frequency shifts were observed when both salt supplemented immobilization buffers were used in detriment of IB PBS, which suggests a higher efficiency in probe immobilization using buffers supplemented with salts. Although higher frequency shifts were obtained using IB NaCl, the results show a higher variation between replicates in comparison to those performed using IB MgCl₂.

Both supplemented buffers were used in the following experiments, in order to further assess their effect on the biosensor hybridization efficiency, in an attempt to establish which buffer should be implemented in an optimized biosensor protocol.

3.3. Effect of immobilization buffer and hybridization temperature on biosensor specificity

Probe immobilization and target hybridization are two highly interdependent aspects in biosensing. If an efficient probe immobilization is not achieved on the biosensing surface, hybridization will not be optimal, and the detection process will be fallible. Considering this, four different experiment sets were conducted in individual electrodes where TCEP-Probe was immobilized using buffers IB NaCl and IB MgCl₂. The specificity of the biosensor was tested in these experiments at two temperature settings, by performing two consecutive hybridization steps using NC Target 1 MM as negative control and C Target as positive control, with washing steps in between (Fig. 4).

The patterned bars in Fig. 4 show the results obtained at RT ($\sim 25^\circ\text{C}$). After TCEP-Probe immobilization in IB NaCl, the Δf values obtained for NC Target 1 MM and C Target (grey patterned bars) were approximate and very inconsistent between replicates, with high standard deviations (-17 ± 19 Hz and -23 ± 17 Hz, respectively), which undermines the specificity and reproducibility of the biosensor.

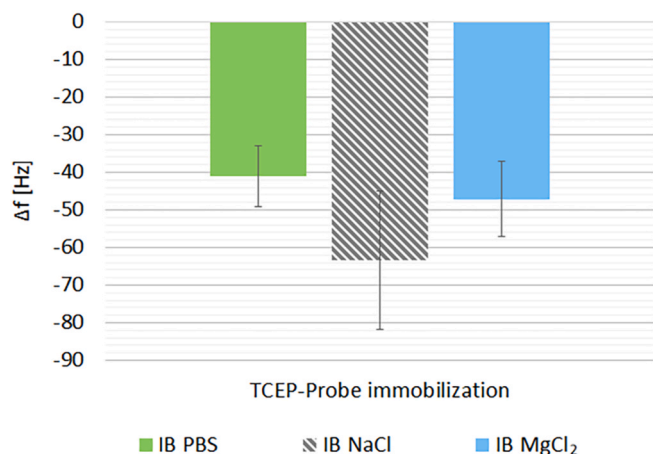


Fig. 3. Frequency shifts Δf (Hz) obtained in the experiments conducted to compare the effect of immobilization buffer (IB) NaCl and IB MgCl₂ in probe immobilization. The solid green bar represents the Δf resultant from TCEP-Probe immobilization in IB PBS (1x PBS); The patterned grey bar represents the Δf resultant from TCEP-Probe immobilization in IB NaCl (1x PBS + 10 mM NaCl); The solid blue bar represents the Δf resultant from TCEP-Probe immobilization in IB MgCl₂ (1x PBS + 1 mM MgCl₂); The error bars show the standard deviation considering three independent immobilizations for each experiment.

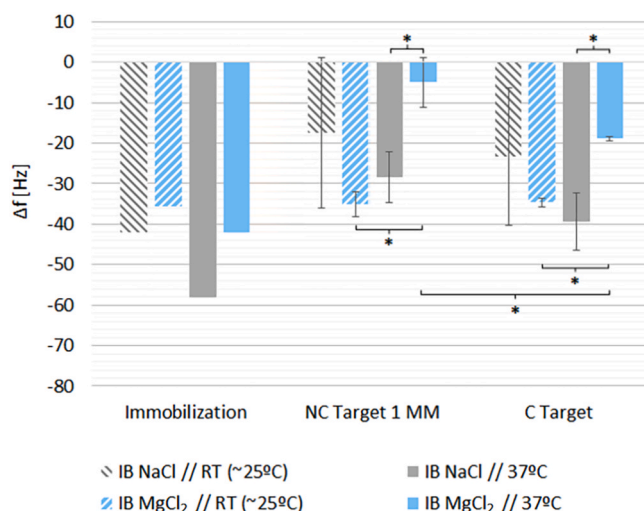


Fig. 4. Frequency shifts Δf (Hz) obtained in the experiments conducted to compare the effect of immobilization buffer and temperature in biosensor specificity. NC Target 1 MM was used as negative control in the experiments and presented a single mismatch in relation to the C Target. The patterned grey bars represent the Δf obtained throughout the experiment where IB NaCl was used for probe immobilization at room temperature (RT $\sim 25^\circ\text{C}$). The solid grey bars represent the Δf obtained throughout the experiment where IB NaCl was used for probe immobilization at 37°C . The patterned blue bars represent the Δf obtained throughout the experiment where IB MgCl₂ was used for probe immobilization at RT $\sim 25^\circ\text{C}$. The solid blue bars represent the Δf obtained throughout the experiment where IB MgCl₂ was used for probe immobilization at 37°C . All hybridization steps were performed in 1x PBS. The error bars show the standard deviation considering three experimental replicates. *Statistically significant signal for a p-value < 0.05 .

When IB MgCl₂ was used for probe immobilization (blue patterned bars), the Δf obtained for NC Target 1 MM (-35 ± 3 Hz) and C Target (-35 ± 1 Hz) were both very similar to the shift resultant from immobilization (-36 Hz). The results suggest the occurrence of hybridization between the NC Target 1 MM and the probe, that became completely unavailable to hybridize with the C Target. In comparison to the previous experiments, the biosensor response was much more consistent, considering the lower standard deviations obtained, but still show lack of detection specificity. Among other factors, hybridization specificity is influenced by temperature. At low temperatures hybridization is more permissive, so mismatched sequences may hybridize. By increasing temperature, only perfectly matches between sequences will be obtained (Zhang, Chen, & Yin, 2012). To improve biosensor specificity and discrimination efficiency, the previous two experiments using IB NaCl and IB MgCl₂ were repeated under a temperature-controlled environment of 37°C (Fig. 3; solid bars) and results were compared with those obtained at RT ($\sim 25^\circ\text{C}$).

When probe immobilization was performed in IB NaCl at 37°C (solid grey bars), less variable frequency shifts were observed during the hybridization steps (-28 ± 6 for NC Target 1 MM and -39 ± 7 for C Target), compared to those obtained at RT (patterned grey bars). The Δf for the C Target at 37°C , although not statistically significant, was greater than the shift for the NC Target 1 MM, suggesting a slight improvement in specificity associated with the temperature increase.

When probe immobilization was performed in IB MgCl₂ buffer, a statistically significant decrease in the Δf for the NC Target 1 MM was observed at 37°C (-5 ± 6 Hz) compared to hybridization with that target at RT (-35 ± 3 Hz). A frequency shift of -19 ± 1 Hz was registered for C Target at 37°C , which was statistically significant in relation to the shift registered for the same hybridization step at RT, and also to the shift registered for the NC Target 1 MM hybridization step performed at 37°C .

Regardless of the immobilization buffer used initially, the

temperature increase improved the hybridization specificity of the biosensor, at different extents. Nevertheless, this effect was more noticeable in the experiments using IB MgCl_2 for probe immobilization, where a higher Δf was achieved for C Target hybridization in detriment to the NC Target 1 MM, and a more stable and reproducible response was achieved considering the experimental replicates. Overall, our results show that the temperature increase to 37 °C hindered hybridization between a single mismatched target and the probe, greatly improving specificity and demonstrating the feasibility of this biosensor for SNP detection. Our findings are in agreement with the research carried out by Song et al. (2022). These authors performed several experiments in a QCM to investigate the temperature effect on the resonant frequency. Among the conducted experiments, the authors explored the temperature effect (ranging from 25 °C to 55 °C) in SNP detection using a thiolated probe immobilized onto a QCM and reported that an increased working temperature enhances the quartz resonant frequency, which in turn improves the sensitivity and accuracy of the functionalized QCM.

The results obtained up to this point allowed us to establish an optimized protocol that involved the immobilization of TCEP-probe in IB MgCl_2 buffer and target/sample hybridization in 1x PBS, in a controlled temperature of 37 °C.

3.4. Real sample analysis – wine authenticity case study

To explore the biosensor performance in real sample analysis, three experiment sets were conducted in individual electrodes, under the pre-established conditions using DNA samples extracted from grapevine varieties Rufete and Touriga Franca. Touriga Franca is complementary to the probe sequence used in the biosensor, while Rufete is heterozygous for two adjacent positions at the probe 5' end, meaning that at each of these two positions, two alleles represented by different nucleotides could be present (Table 1).

Depending on what allele is present, hybridization between this sample may or may not occur. The tested DNA samples also differed in their complexity. DNA was extracted from leaf and experimental wine samples of the two varieties. Leaf DNA is characterized by high purity, low presence of contaminants and high integrity, while wine DNA is heavily contaminated with different compounds, such as phenolic compounds, and is highly degraded.

In each experiment set, TCEP-probe was immobilized in the electrode, followed by sequentially testing Rufete DNA and Touriga Franca DNA samples, with washing steps in between. Sample testing steps were performed in triplicate as it follows: after hybridization with a complementary sample, the probe/sample complex was reversed by denaturation, so further hybridizations could occur in the same functionalized electrode. The results are presented in Fig. 5.

The solid green bars represent the frequency shifts obtained during experiments using leaf DNA samples. TCEP-probe immobilization resulted in a frequency shift of -80 Hz. When tested, Rufete leaf DNA

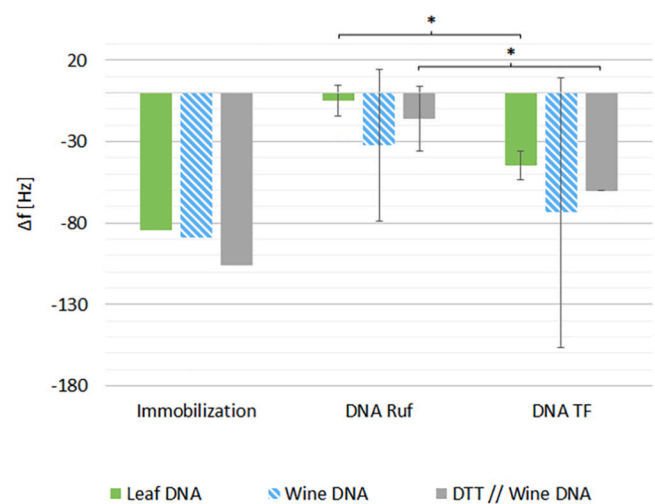


Fig. 5. Frequency shifts Δf (Hz) obtained in the experiments conducted to test the biosensor performance in real sample analysis (leaf and wine extracted DNA). Samples from two grapevine varieties were tested. Regarding the 35 bp probe sequence, variety Rufete is heterozygous in two mismatched positions (negative sample) while Touriga Franca is fully complementary to the probe (positive sample). The solid green bars represent the Δf obtained throughout the experiment set using leaf DNA. The patterned blue bars represent the Δf obtained throughout the experiment set using wine DNA. The solid grey bars represent the Δf obtained throughout the experiment set using DTT after probe immobilization and wine DNA samples. All experiments were conducted at 37 °C using IB MgCl_2 during probe immobilization. All hybridization steps were performed using 1x PBS. The error bars show the standard deviation of three experimental replicates. *Statistically significant signal for a p-value < 0.05.

sample resulted in a Δf of -5 ± 9 Hz, which is negligible in comparison to the Δf obtained from probe immobilization. When testing Touriga Franca leaf DNA, a Δf value of -45 ± 9 Hz was observed, which was statistically significant in relation to the Δf of the Rufete sample. Throughout the experimental replicates, the biosensor response was reproducible and consistent. This shows that hybridization only occurred between the probe and the fully complementary Touriga Franca DNA, and that the biosensor was able to effectively detect the presence of the complementary sequence in leaf DNA samples. The lack of hybridization between Rufete DNA and the probe also demonstrates the specificity of the biosensor, that could discriminate two sequences presenting only two mismatches.

The results from the next experiment set are represented in blue patterned bars. The same protocol was followed as before, except this time wine DNA samples were tested. Probe immobilization was successful, and a Δf value of -89 Hz was registered. In the hybridization steps, a Δf of -32 ± 47 Hz was registered upon Rufete wine DNA analysis, followed by -74 ± 83 Hz for Touriga Franca. In comparison to the biosensor response when leaf DNA samples were tested, in wine DNA a higher response was achieved in the analysis of both samples, but those were highly unstable and variable throughout replicates. As the complexity of tested samples increased significantly in this experiment (Table 1), these variations could be explained by possible interactions between the contaminant compounds present in wine DNA samples and the immobilized probe, and/or the gold electrode surface left vacant after probe immobilization. This would have led to inflated frequency shifts caused by contamination-induced mass deposition. As described by Oberhaus, Frense, and Beckmann (2020), amino acids, proteins and phenols are examples of fouling agents that have the capacity of adhering to gold surfaces through different non-specific binding mechanisms, namely adsorption, polymerization, and precipitation. This causes misleading signal readings and leads to a layer formation at the substrate surface, that inhibits the interactions between the immobilized probe with the target analyte and hampers the detection process. By

Table 1

Sample information regarding the mismatches present at each variety in relation to the probe sequence, the DNA yield and absorbance ratios at 260nm/280 nm obtained from the spectrophotometric analysis of DNA extracted from leaf and wine samples. An A_{260}/A_{280} value ~ 1.8 is generally accepted for pure DNA. A lower ratio indicates presence of contaminants, such as phenols.

Mismatched sites	Leaf DNA samples quantification				Wine DNA samples quantification	
	Probe (5' – 3')	A	A	DNA yield (ng/μL)	DNA yield (ng/μL)	A_{260}/A_{280}
Touriga Franca (complementary)	T	T	111.36	1.73	310.85	1.40
Rufete (two mismatches)	W	W	83.36	1.77	338.59	1.21

Nucleotide code: W – A or T.

preparing the gold surface through antifouling strategies that block this non-specific mass deposition, an improvement in sensitivity, reproducibility, and overall robustness of the biosensor should be attained regarding practical applications in complex samples. Dithiothreitol (DTT) can strongly and rapidly adsorb to gold surfaces via its two sulfhydryl groups, and thus block the vacant space after probe immobilization. Its function as an antifouling agent in biosensor development has been demonstrated in configurations using gold electrodes (Lino et al., 2023; Liu & Wang, 2019; Wang, Dong, Liu, & Wang, 2015) and gold nanoparticles (Liang, Canoura, Yu, Alkhamis, & Xiao, 2018).

To improve biosensor performance and reproducibility in wine DNA analysis, a third experiment set was performed using a 12.5 μM DTT solution added immediately following probe immobilization. The results are represented by solid grey bars in Fig. 5. A frequency shift of -110 Hz was observed for probe immobilization and surface blocking by DTT. Upon Rufete wine DNA testing, a Δf of -16 ± 20 Hz was registered. In comparison to what was observed for this sample in the previous experiment, a lower Δf was registered, and a considerably more consistent response was observed throughout the replicates. When testing the complementary Touriga Franca sample, a higher statistically significant frequency shift was registered (-60 ± 0 Hz) in comparison to that obtained upon analysis of the mismatched Rufete sample. This response was also consistent throughout the experiment replicates. These results prove that the blocking step performed by DTT had a positive effect in avoiding wine sample contaminants deposition on the electrode, substantially improving biosensor performance in detecting the complementary sequence present in complex samples. The results obtained from this last experiment were approximate to those obtained from leaf DNA analysis and prove the efficiency of the biosensor, not only in detecting targeted DNA in samples with different characteristics, but also its high detection specificity when discriminating samples at mismatch level using a simple protocol.

In both experiments performed using leaf DNA and the optimized protocol in wine DNA, no significant hybridization was observed between Rufete DNA and the probe.

These results suggest that Rufete samples presented a heterozygous haplotype (AT or TA) regarding the two sequence positions where mismatches were identified. Otherwise, if Rufete presented a homozygous haplotype (AA or TT), complementary sequences would be present in half concentration in the samples from this variety in comparison to those of Touriga Franca, and an increased frequency shift would have been observed, resultant from hybridization between this sample and the probe. This is similar to what was observed in a previous study performed by our group (Barrias, Fernandes, Eiras-Dias, Brazão, & Martins-Lopes, 2019). In that study, an optical DNA-based biosensor was applied to the analysis of must and wine DNA samples from grapevine varieties Malvasia Fina, Tinta Amarela and Tinta Roriz. Malvasia Fina was heterozygous at three positions within the sequence targeted by the probe, which was the exact same probe sequence used in the current study to develop a QCM DNA-based biosensor. The results from that study showed the occurrence of hybridization between the probe and Malvasia Fina samples, suggesting that in that case, the samples tested from that variety presented a haplotype that was complementary to the probe, present in half the concentration in comparison to the complementary variety Touriga Franca. To confirm the haplotype present in the Rufete variety, (and other heterozygous varieties) the two homologous chromosomes concerning the SNPs targeted by the probe would need to be sequenced.

Another aspect to consider is that the SNPs within the probe sequence are adjacent, and adjacent SNPs tend to be inherited together (Lee et al., 2017). By consequence, these might provide redundant information in terms of genetic variation and might also not capture enough genetic diversity to reliably differentiate between varieties. To surpass this and increase the biosensor discriminative power, different probes targeting other distant SNP markers may be immobilized onto several electrodes in a multi-QCM setup. This could even be

implemented in a microfluidic configuration, allowing for a unique sample to pass through the multiple QCMs, so different SNPs could be detected.

Regarding other QCM-DNA biosensors, the majority was developed for clinical applications and involve more intricate protocols than those implemented in our approach. For instance, Ma et al. (2020) developed a QCM-DNA biosensor for the detection of miR-221, a short noncoding RNA identified as a biomarker in different types of cancer. Besides typical QCM instrumentation, the functioning principle behind the biosensor was dependent on the immobilization of a biotinylated DNA probe, and the application of streptavidin/metallic-organic frameworks as well as exponential isothermal amplification for signal amplification and enhanced sensitivity. The authors also adopted a more elaborate QCM cleaning procedure and longer incubation periods. The choice for a more complex protocol is understandable, considering that the device is aimed at clinical applications in more strict and precise settings, where trace detection of target molecules is peremptory. In another field of application, Wang et al. (2017) reported a QCM-DNA biosensor aimed at detecting the H5N1 avian influenza virus. The authors prepared a nanoporous gold film that was deposited onto the gold electrode on the QCM surface, to form a nanowell-based electrode that served as a substrate for DNA probe immobilization, which was considerably improved in comparison to direct immobilization into an unmodified QCM. Although several processes and compounds were required to construct the sensing interface on the biosensor, the enhancement of the immobilization step had a positive effect on the detection process, that was achieved in 10 min using stock H5N1 solutions and chicken tracheal swab samples spiked with H5N1. Ozalp, Bayramoglu, Erdem, and Arica (2015) proposed a QCM-DNA biosensor for pathogen detection, demonstrating the device practical applicability to the detection of *Salmonella typhimurium* in milk samples. Once again, the biosensing protocol was more elaborated to the one herein. They combined magnetic beads with the QCM typical instrumentation to integrate a purification step within the biosensor, that involved thorough preparation and functionalization of the beads, an abrasive gold surface cleaning protocol, and two steps of probe immobilization onto different substrates. Nevertheless, the authors report a rapid detection time of 10 min, which is desirable for food testing performed by biosensing analysis.

4. Conclusion

In our study, a QCM-DNA biosensor was developed for SNP detection in samples of varying complexity. To achieve this, a protocol was optimized by performing several experiments focusing on different parameters affecting probe immobilization and hybridization efficiency. We were able to show that the reduction of the thiolated probe with TCEP improves the immobilization of the probe, and this process is also improved in the presence of a PBS buffer supplemented with MgCl_2 . The specificity of the biosensor was also demonstrated at 37°C , as we were able to detect a single mismatch with synthetic oligonucleotides. The biosensor was used in a real sample analysis for authenticity assessment of wine, where the addition of DTT as an antifouling agent improved the stability and reproducibility of the biosensor when in contact with highly complex samples. The QCM-DNA biosensor shows great potential for practical applications as it enabled the specific detection of complementary DNA in both leaf and wine DNA samples, to the detriment of samples in which two heterozygous mismatches were present. Considering the characteristics of wine DNA (highly fragmented, degraded and contaminated), the biosensor herein proposed can potentially be an efficient and reliable alternative to PCR-based methodologies. Regarding wine authenticity, the application of these approaches in practical contexts is still constrained, mainly due to the size of fragments present in wine samples being below the required. Furthermore, the established biosensor depends only on DNA extraction, simple instruments and reagents and does not require PCR amplification steps or

further signal amplification strategies. The simplicity of the implemented protocol, along with the results obtained in our work, demonstrates the potential of the developed QCM-DNA biosensor in wine authenticity assessment.

CRediT authorship contribution statement

Sara Barrias: Writing – original draft, Visualization, Investigation, Formal analysis. **José R. Fernandes:** Writing – review & editing, Resources. **Paula Martins-Lopes:** Writing – review & editing, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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